Measurement of distance

Task: Measure a distance with the HC-SR04 ultrasonic sensor and show the result on the serial monitor.

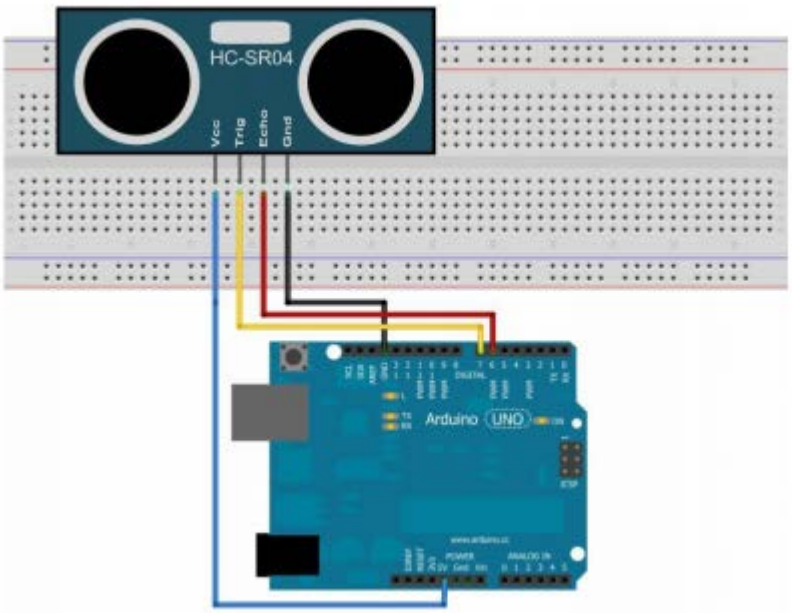
Required equipment: microcontroller board / cables / Breadboard / HC-SR04 ultrasonic sensor

How does the ultrasonic sensor work?

The sensor has four pins.

a) 5V (+) b) GND (-) c) echo d) trigger

The contacts 5V and GND are meant for the power supply. Trough the "trigger" pin the sensor gets a short signal (5V) from the microcontroller board to create a sound wave. As soon as the sound wave hits a wall or other objects, it will be reflected and comes back to the ultrasonic sensor. When the sensor detects this returned sound wave, the sensor will send a signal to the Arduino microcontroller trough the "echo" pin. The Arduino-board measures the time between the transmission and the return of the sound wave, and converts this time into a distance.



Code:

```
int trigger=7;  //"trigger" on pin 7.

int echo=6;   //"echo" on pin 6.

long time=0;   //The value "time" will safe the time between transmission and
//returning of the soundwave.

long dist=0;  //The value "dist" will save the calculated distance. It will
//start with "0". Instead of "int" we are using "long" for this value, to save a
//bigger number.
```

```
 void setup()

 {

 Serial.begin (9600);   //Starting the serial communication. It will send the
 //data from the arduino board to the computer to show it on the serial monitor.

 pinMode(trigger, OUTPUT);    //"trigger" (Pin 7) is an output.

 pinMode(echo, INPUT);  //"echo" (Pin 6) is an input.

 }


 void loop()

 {

 digitalWrite(trigger, LOW);   //Low voltage on the trigger pin to produce a
 //clear signal.

 delay(5);   //….for 5 milliseconds.

 digitalWrite(trigger, HIGH);    //Creating the soundwave.

 delay(10);   //..for 10 milliseconds.

 digitalWrite(trigger, LOW);  //Stop creating the soundwave.


time = pulseIn(echo, HIGH);  //With the command pulseIn (Capital "i" in the
//front of the "n") the arduino board measures the time between sending and
//receiving the soundwave.

dist = (time/2) / 29.1;  //This calculation transforms the measured time into
//the distance in centimeter. (The sound needs 29,1 seconds for one centimeter.
//The time gets divided with two, because we only want to get one distance and
//not the two ways that the soundwave has to take).
```

```
if (dist >= 500 || dist <= 0)   //If the distance gets over 500cm OR under 0cm,
//the measurement is no longer accurate.

{

Serial.println("No measurement");  //So the serial monitor displays “No
//measurement”
}

else   //otherwise
{

Serial.print(dist);  //The calculated distance is shown on the serial monitor.

Serial.println("cm");
}

delay(1000);  //This command causes a short break between the measurements.
}
```